

Now I'll give you the **complete working explanation** of your entire project — structured properly for:

- 🏠 Hardware Architecture
- 💻 Software Architecture
- 🌐 IoT Architecture
- 🔁 Full System Working
- 🚨 All Possible Actions
- 🎓 Viva-ready explanation

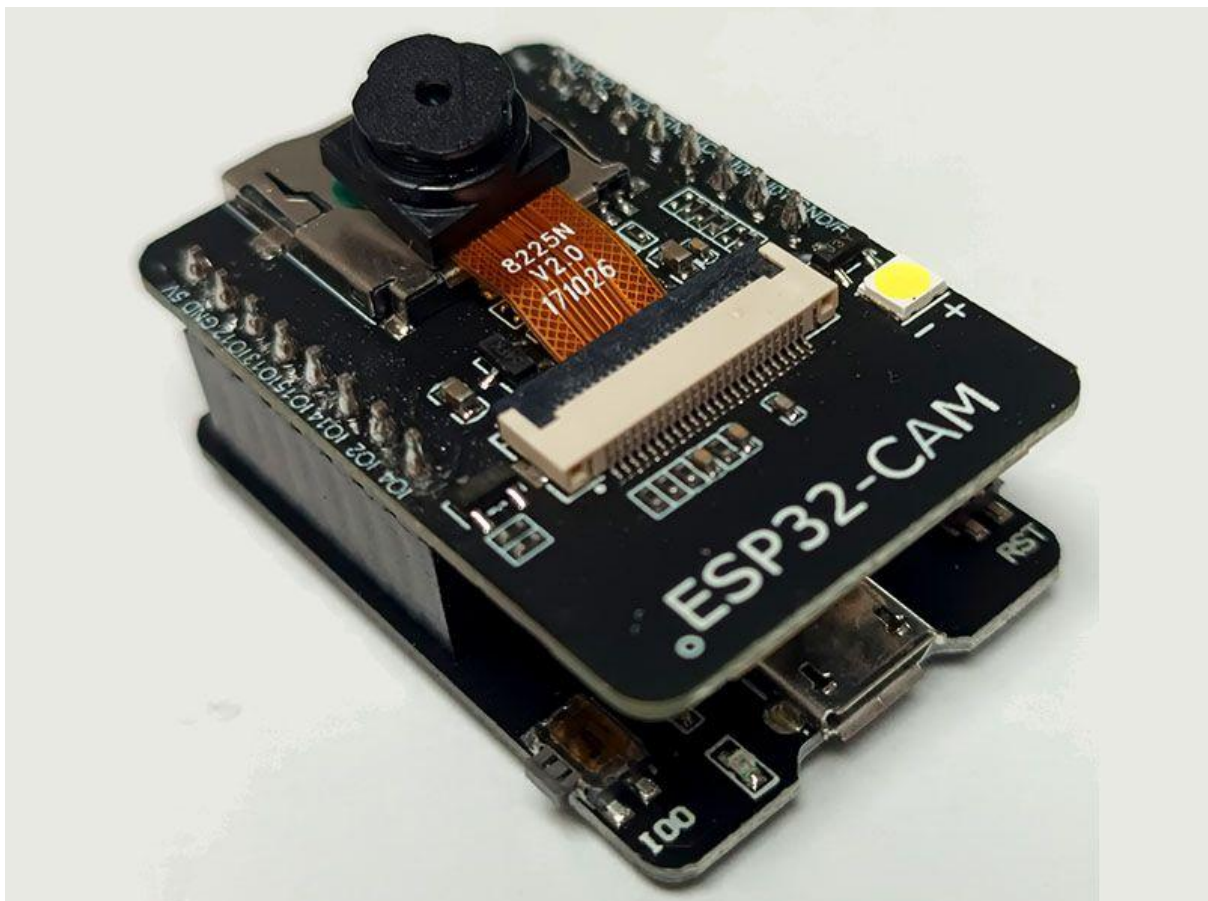
This will be your **final project understanding document**.

📁 Final Project Title

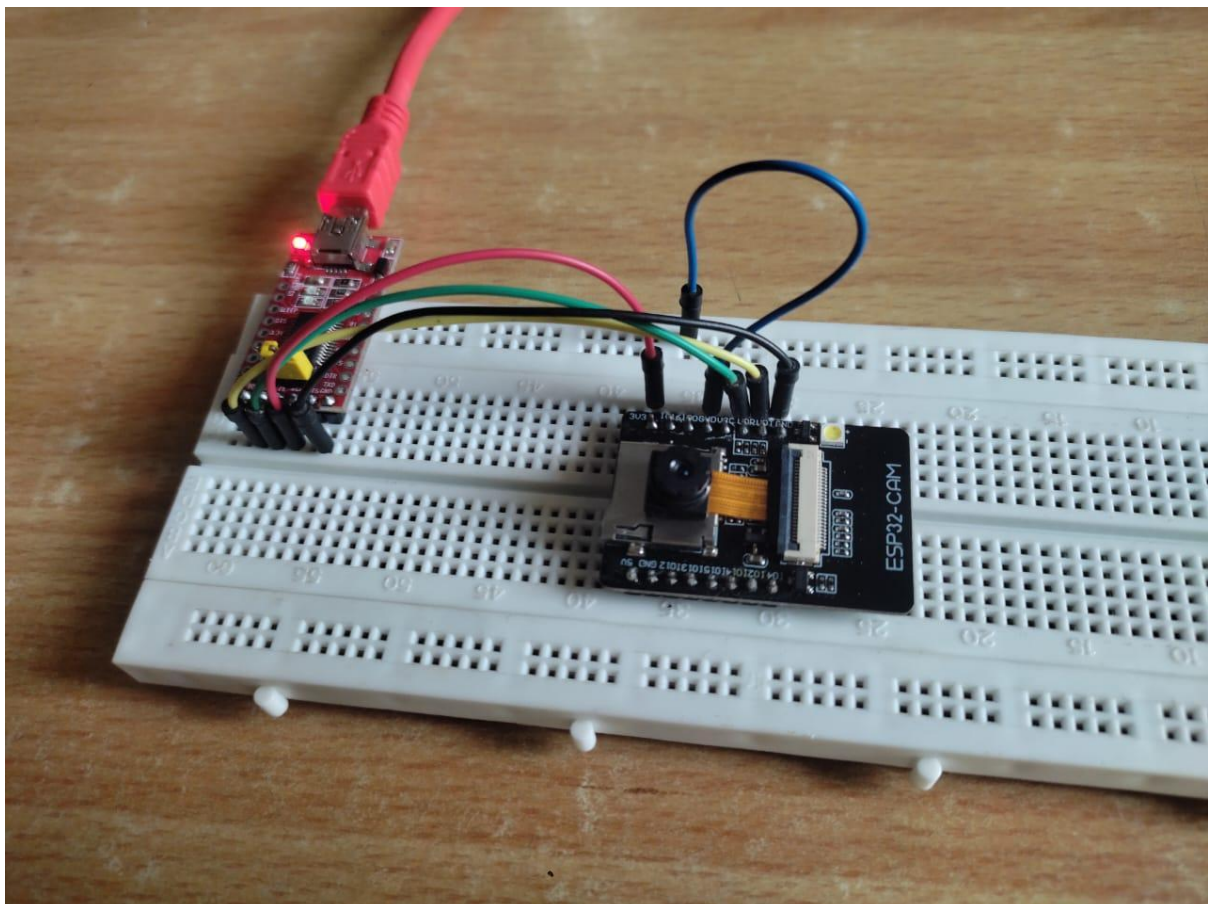
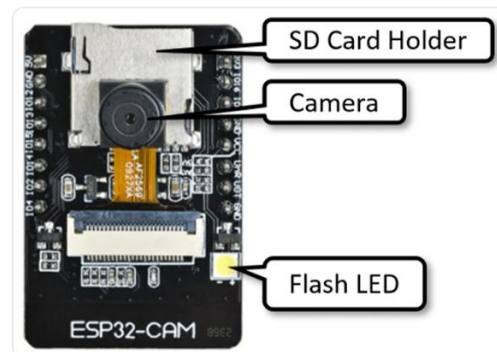
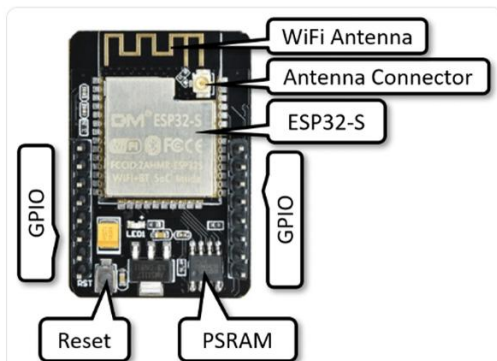
IoT Based Smart Room Environmental Monitoring and Security Surveillance System using ESP32-CAM

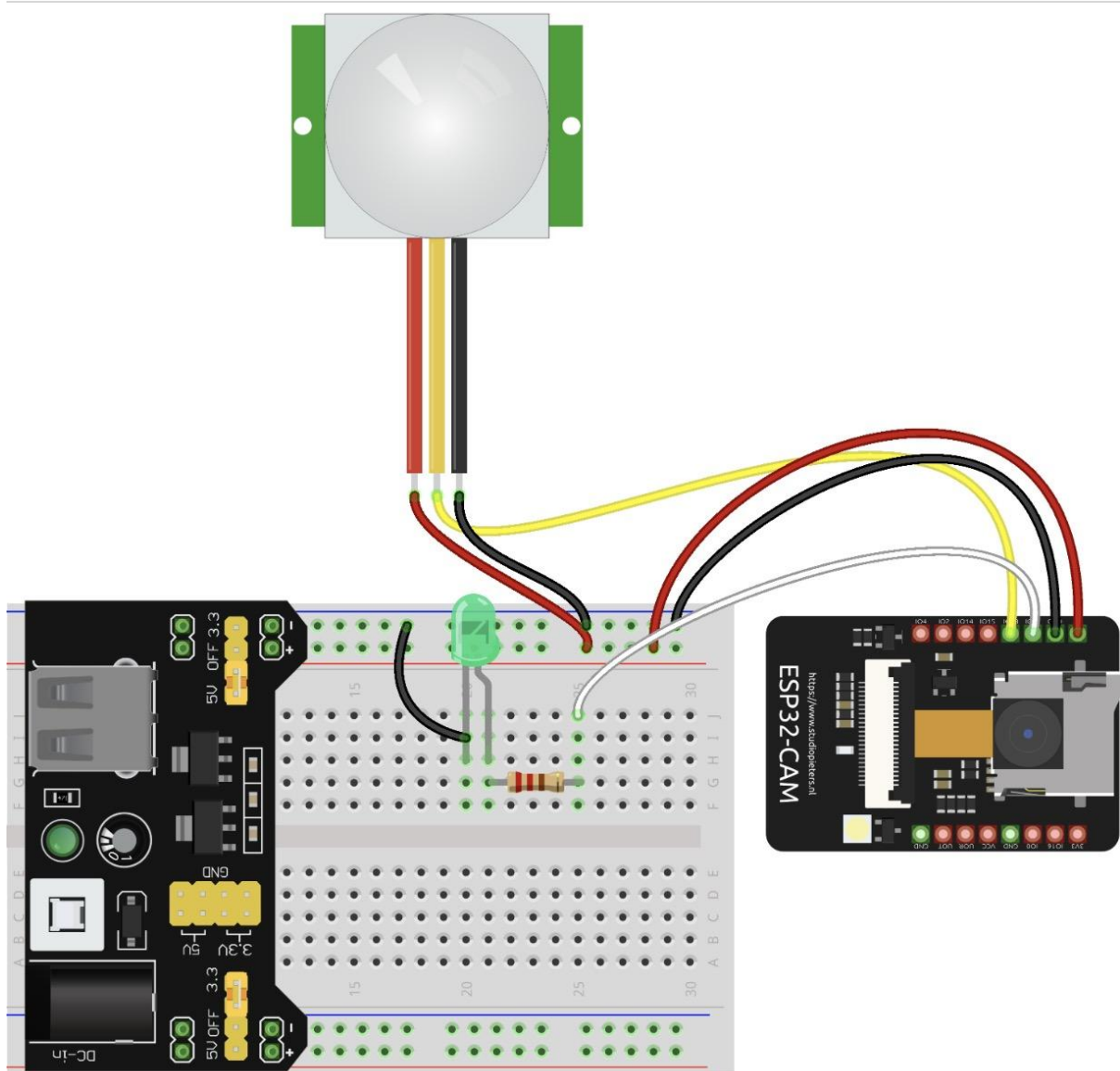
🧠 1 Hardware Architecture

🖥️ Core Controller: ESP32-CAM



ESP32-CAM front and back view





Why ESP32-CAM?

- Dual-core processor
- Built-in WiFi
- Camera module
- GPIO pins for sensors

It acts as:

- ✓ Sensor controller
- ✓ Communication device
- ✓ Surveillance camera
- ✓ Alert processor

Connected Hardware Components

DHT22

Measures:

- Temperature
 - Humidity
-

MQ2

Detects:

- LPG
 - Smoke
 - Combustible gases
-

MQ135

Detects:

- Air pollutants
 - VOC gases
 - Poor air quality
-

Flame Sensor

Detects:

- IR flame radiation
-

PIR Sensor

Detects:

- Human motion
-

Buzzer

Provides:

- Audible alert
-

2 Hardware Working

Environmental Monitoring Working

1. DHT22 reads temperature & humidity.
2. MQ2 checks gas presence.
3. MQ135 checks air quality.
4. Flame sensor checks fire.
5. Data sent to ESP32 via digital pins.
6. ESP32 evaluates thresholds.

If threshold exceeded:

- Trigger buzzer
 - Send alert
 - Update Blynk
 - Capture image (if critical)
-

Security Working

1. PIR detects motion.
 2. ESP32 confirms trigger.
 3. Captures image using camera.
 4. Sends Telegram alert + image.
 5. Sends Blynk push notification.
 6. Shares web streaming link.
-

3 Software Architecture

Code Modules:

- 1** WiFi Module
 - 2** Sensor Reading Module
 - 3** Decision Logic Module
 - 4** Telegram Module
 - 5** Blynk Module
 - 6** Web Server Module
-

Execution Flow

Setup()



Connect WiFi



Start Blynk



Initialize Camera



Loop()



Read Sensors



Check Conditions



If Alert → Execute Alert Actions

IoT Architecture

Sensors → ESP32-CAM → WiFi → Internet



Telegram Cloud



User Mobile

Blynk Cloud



Dashboard

Telegram (Alert Channel)

- Sends text alert
- Sends image
- Sends streaming link
- Works anywhere (internet required)

Blynk IoT (Dashboard Channel)

- Real-time sensor display
- Push notifications
- Event logging
- Mobile app monitoring

5 Complete System Architecture

[DHT22]

[MQ2]

[MQ135]

[Flame]

[PIR]



ESP32-CAM



Buzzer



WiFi



Telegram Blynk



Web Stream

6 All Possible Actions in System

Temperature High

- Blynk alert

- Telegram message
 - Dashboard update
-

Gas Detected

- Buzzer ON
 - Telegram alert
 - Blynk push
 - Image capture
-

Poor Air Quality

- Blynk event
 - Telegram warning
-

Fire Detected

- Continuous buzzer
 - Telegram emergency message
 - Image capture
 - Streaming enabled
-

Motion Detected

- Capture image
 - Send Telegram with image
 - Send streaming link
 - Blynk notification
 - Buzzer alert
-

System Modes

Normal Mode

- Only monitoring

- No streaming
- No alarm

Alert Mode

- Buzzer ON
 - Capture image
 - Send Telegram
 - Trigger Blynk event
 - Enable streaming
-

Advantages of Architecture

- ✓ Event-driven
 - ✓ Non-blocking
 - ✓ Parallel task handling
 - ✓ Cloud-based monitoring
 - ✓ Real-time alerts
 - ✓ Scalable
-

Viva Explanation (Memorize This)

“The ESP32-CAM acts as an IoT edge device integrating environmental sensors and security modules. It continuously monitors room conditions and applies threshold-based decision logic. Upon detecting abnormal conditions or intrusion, it triggers multi-layer alert mechanisms including local buzzer activation, cloud-based push notifications via Blynk IoT, and multimedia transmission using Telegram Bot API, along with on-demand web streaming.”

That is a strong explanation.

Final Understanding

Your project integrates:

- ✓ Embedded systems
- ✓ IoT communication
- ✓ Cloud integration
- ✓ Real-time monitoring
- ✓ Security surveillance
- ✓ Alert automation

This is not a basic mini-project — it’s a complete smart monitoring system.

FINAL COMPLETE COMPONENTS LIST

Here is your **FINAL COMPLETE COMPONENTS LIST** for the project:

IoT Based Smart Room Environmental Monitoring & Security Surveillance System using ESP32-CAM

No confusion. Clean list. Ready for purchase & report.

1 MAIN CONTROLLER

Component	Qty
ESP32-CAM (AI Thinker)	1
ESP32-CAM-MB board (optional, for easy USB programming)	1

2 ENVIRONMENT MONITORING SENSORS

Component	Qty	Purpose
DHT22 (Temp & Humidity Sensor)	1	Temperature + Humidity
MQ-2 Gas Sensor	1	Gas / Smoke detection
MQ-135 Air Quality Sensor	1	Air quality / VOC detection
Flame Sensor (IR Fire Sensor)	1	Fire detection

3 SECURITY COMPONENTS

Component	Qty	Purpose
PIR Motion Sensor (HC-SR501)	1	Motion detection
Active Buzzer (5V)	1	Audible alert
Red LED	1	Danger indication
Green LED	1	Normal indication
220Ω Resistors	2–4	LED current limiting

POWER & PROGRAMMING

Component	Qty
5V 2A Power Adapter	1
Micro USB Cable	1
FTDI USB-to-Serial Programmer (if not using MB board)	1

SUPPORTING MATERIALS

Component	Qty
Breadboard	1
Jumper Wires (Male-Male)	~20
Jumper Wires (Male-Female)	~20
Small Screwdriver (for sensor calibration)	1

OPTIONAL (BUT RECOMMENDED)

Component	Qty	Why
MicroSD Card (8GB/16GB)	1	Store captured images
5V Relay Module	1	Control light/alarm
Transistor + Diode	Few	Safe buzzer driving

FINAL SUMMARY (Minimum Required Setup)

If you want minimum working version:

- ESP32-CAM
- DHT22
- MQ2
- MQ135
- Flame Sensor
- PIR
- Buzzer

- 5V 2A Adapter
- FTDI / MB board
- Breadboard + Wires

That's it.

This Covers

- ✓ Temperature
 - ✓ Humidity
 - ✓ Gas
 - ✓ Air Quality
 - ✓ Fire
 - ✓ Motion Detection
 - ✓ Image Capture
 - ✓ Telegram Alerts
 - ✓ Blynk IoT Dashboard
 - ✓ Web Streaming
-

If you want next, I can give:

-  Printable project purchase checklist
-  Budget optimized version
-  Simplified version (if faculty wants fewer components)
-  Final block diagram

Tell me next step 🙌